



9/1/2026

N + - Network Fundamentals & Building Networks.

Introduction to Networking and the OSI
Model.

TASK 1.

❖ **Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.**

➤ **Wireshark Packet Capture:**

Procedure:

- Installed **Wireshark** on the laptop.
- Opened Wireshark and selected the **main network interface** (Wi-Fi/Ethernet).
- Started the packet capture.
- Opened a web browser and visited **Instagram.com**.
- Stopped the capture after the website loaded.
- Checked the **Protocol** column in the packet list.

Protocol Names Observed:

Protocol
DNS
TCP
TLS
HTTP
ARP
ICMP

TASK 2.

❖ Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.), Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

➤ **Mapping Wireshark Packets to OSI Layers:**

I selected three packets from the Wireshark capture and mapped them to their main OSI layers using the Protocol and Info details.

Packet	Protocol	Info / Purpose	OSI Layer
1	DNS	Resolves the website name into an IP address	Layer 7 – Application
2	TCP	Provides reliable data transfer between devices	Layer 4 – Transport
3	TLS	Secures the communication by encrypting data	Layer 6 – Presentation

TASK 3.

- ❖ Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

➤ OSI Model Diagram:

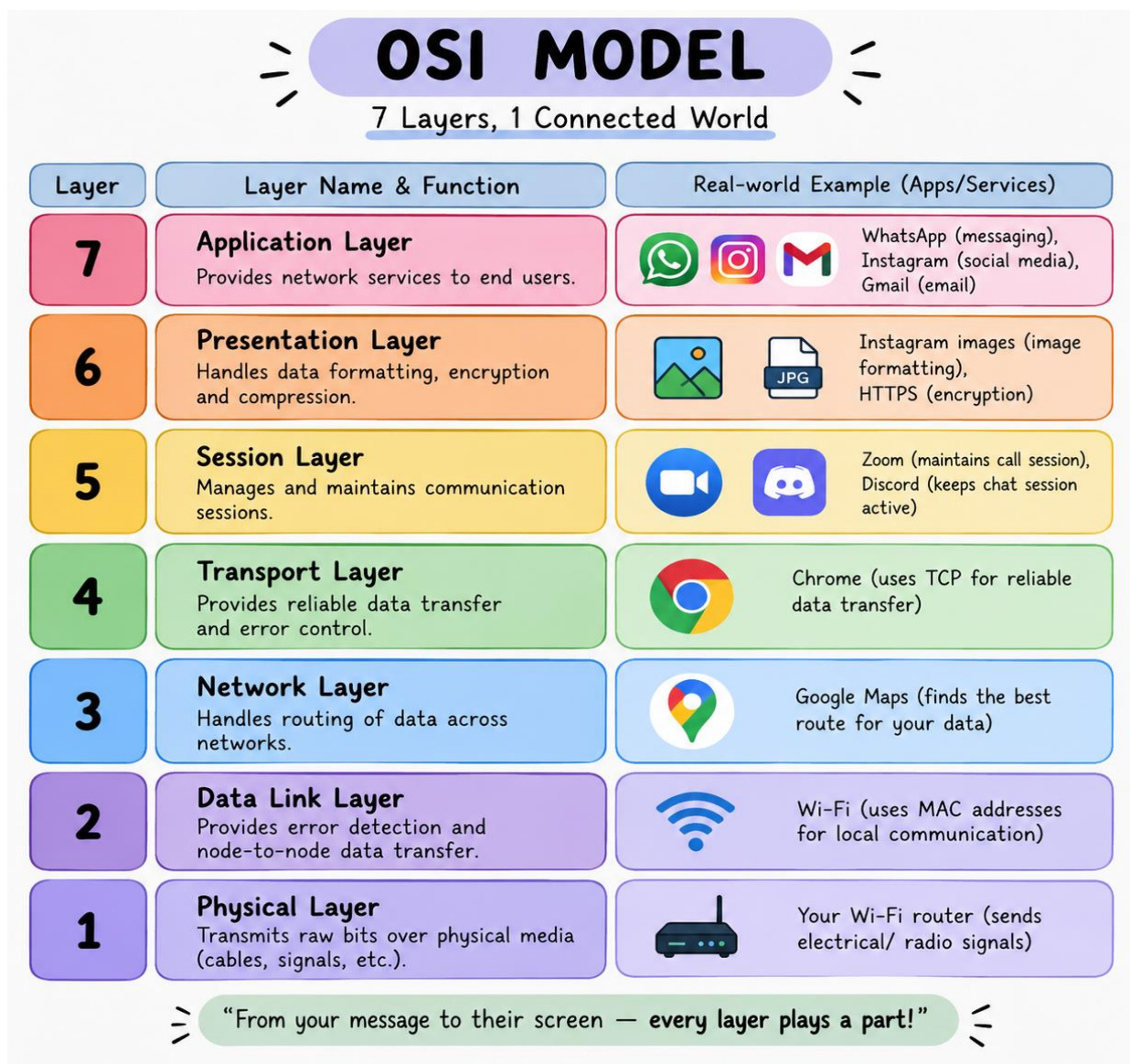


Figure : OSI Model with Real-World Examples.

Layer	OSI Layer	Real-World Example
7	Application	WhatsApp
6	Presentation	Instagram Images
5	Session	Zoom
4	Transport	Chrome
3	Network	Google Maps
2	Data Link	Wi-Fi
1	Physical	Wi-Fi Router

TASK 4.

❖ **Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words, Think about how your message is typed, packaged, sent over the internet, and received.**

➤ **WhatsApp Message Through the OSI Model:**

When I send a message on WhatsApp, the data passes through different OSI layers before reaching the receiver.

OSI Layer	What Happens
Layer 7 Application	I type and send the message using WhatsApp. The application prepares the message for communication.

Layer 6 Presentation	The message is formatted and encrypted so that it can be securely transmitted.
Layer 4 Transport	The message is divided into smaller data units and delivered reliably to the receiver.
Layer 3 Network	IP addressing and routing are used to send the data across the Internet toward the receiver.
Layer 2 Data Link	The data is placed into frames and delivered between devices on each local network.
Layer 1 Physical	The data is transmitted as electrical, radio, or optical signals through Wi-Fi, cables, or other physical media.

At the receiver's side, these processes are reversed, and the original message appears in WhatsApp.